

Use Case: ACH Payment Processing Service

Business Context

Your company provides payment infrastructure for enterprise customers.

Customers integrate with your platform via APIs to initiate ACH payments. Once a payment request is submitted, it must be processed asynchronously through banking partners.

Because banking systems are not always available and network failures occur regularly, the platform must be resilient, observable, and capable of safely recovering from failures.

You have been asked to design a backend service responsible for accepting ACH payment requests and managing the payment lifecycle.

Functional Requirements

Payment Submission API

A client should be able to submit an ACH payment request containing:

- Customer identifier
- Source account
- Destination account
- Payment amount
- Payment reference

Example:

```
{  
  "customerId": "C12345",  
  "sourceAccount": "VA10001",  
  "destinationAccount": "EXT98765",  
  "amount": 250.00,  
  "reference": "PMT-1001"  
}
```

Payment Lifecycle

A payment can move through the following states:

```
PENDING  
PROCESSING  
COMPLETED  
FAILED  
RETRYING
```

Asynchronous Processing

Submitted payments should not be processed synchronously.

The system should support:

- Queue-based processing
- External bank API integrations
- Retry handling

Duplicate Requests

Clients may accidentally submit the same payment request multiple times.

The system must prevent duplicate payment execution.

Status Retrieval

Clients should be able to retrieve payment status using an API.

Event Notifications

Clients should be able to receive payment status updates through webhooks.

Audit-ability

All payment state changes must be traceable.

Deliverables

1. HL Architecture / Solution Approach
2. Data Model (if any)
3. API Design
4. Code Implementation